

CS 281 Fall 2025-2026

Project Topics

Carefully study the following project descriptions and try to capture the following:

- Strong and weak entities,
- Relationships (binary, ternary and specialization (IS-A) relationships), role indicators for an entity in a relationship, mapping cardinalities (1-1, 1-M, M-M) and aggregations (i.e., a relationship as a whole serves as an entity in another relationship),
- Primary and Foreign Keys.

REMARK: In each of the scenarios below, we try to provide examples of most of the above E/R elements, so please study these scenarios carefully!

You are free (and indeed *expected to*) add more details (and, thus, entities/relationships) to the scenarios below. While designing your database, also keep in mind that in the upcoming stages of the project, you have to specify interesting/sophisticated queries over this database.

- **Digital Music/Podcast Streaming Service:** In this project, you are asked to design a database for a digital music and podcast streaming service, similar to *Spotify* or *Apple Music*.

Each user has a unique username, password, first name, last name, and country information. There are three types of users: **Admin**, **Artist**, and **Member**. Additionally, Artists have a pseudonym and a description. Artists can create audio tracks, which may be either songs or podcasts. Each audio track has a unique track ID, name, genre, duration in seconds, and release date. Artists can collaborate with other artists to create audio tracks. They can also group songs into an album. An album has a unique album ID, name, and release date. Both Members and Artists can create playlists that consist of songs or podcasts. Similar to albums, playlists also have unique IDs and names. Members can listen to an audio track multiple times and can also follow one another. Audio tracks, albums, or playlists can be liked by Members. Finally, Admins are responsible for approving new Artist sign-ups.

- **Media Sharing Application:** In this project, you are asked to design a database for a Media Sharing Application where users can share images or videos, like *Instagram*.

Each user has a unique username, password, name, and surname. There are two types of accounts: **Admin** and **User**. User accounts can set a profile description and personal information as well. Users can create posts. A post has a unique ID, a release date, a description, and a link. A post can be either a single media or a collage that consists of multiple media. A media medium has a unique ID and resolution. Media can be either a photo or a video. Videos additionally have durations. Different than posts, users can share stories which will only be visible for 24 hours. A story has a unique story number, a link, and content. Users can like the posts of other users, send comments, or tag multiple users in photos or videos. Each comment must be stored with a unique comment ID, sending date, and content. Users can follow each other. They additionally have an account type that can be set to public or private. If a user account is private, only the followers of that user can see their posts. Otherwise, all the information is visible to everyone. Admins are responsible for deleting inappropriate posts.

- **Theatre Ticket Selling System:** In this project, you are asked to design a database for users who want to purchase tickets for their desired theatre plays, such as *Bubilet* or *Biletix*.

Each user has a unique email, password, first name, and last name. There are three types of users: **Admin**, **Customer**, and **Theatre Officer**. For Customers, you should also store their addresses, phone numbers, and credit card information. There will be plays with a unique play number, title, playwright, genre, description, and duration. A play can be performed in different theatres across the country. A theatre has a unique name, address, and seating capacity. Each theatre contains seats identified by row and seat numbers. For each play, there will be multiple stagings. Each staging must have a unique staging number and date, and it must be associated with both the play and the theatre. There are actors who can perform in different plays. Each actor has a first name, last name, gender, height, and age. A Customer can book one or multiple seats for a specific staging with a single booking. After a play inside a staging is over, Customers can leave a comment with a unique number, content, and comment date. Theatre Officers are employees who work in theatres, while Admins are responsible for adding new plays and stagings to the system.

- **House Rental System:** In this project, you are asked to design a data management system for a house/room rental system like *Airbnb*.

Each user has a unique ID, name, and email address. There are two types of users: **Property Owners** and **Customers**. For Property Owners, the system must also store their GSM numbers. Each owner may have multiple properties, but each property belongs to exactly one owner. A property has a unique property number, details such as name, city, district, description, capacity, property type (house/room), and nightly price. For each property, the system must keep availability information. Property Owners can mark specific dates as “not available.” If a property is available, Customers may make a booking for a desired period. Each booking has a unique booking number and includes the property, the customer, the start and end dates, and the total paid price. Once a booking is finalized, the reserved dates are automatically marked as “not available.” After their stay, Customers can leave an evaluation for their booking. Each evaluation has a unique number, a rating (1–5), and an optional comment.